Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

1. Administrative & Study Identification

Full Study Title: Phase 2a Proof-of-Concept, Multicenter, Randomized, Open Label Study to Evaluate the Efficacy, Safety, and Pharmacokinetics of a Single Dose of the Combination M5717-pyronaridine as Chemoprevention in Asymptomatic Adults and Adolescents With Plasmodium Falciparum Malaria Infection (CAPTURE-2) **Short Title/Acronym:** CAPTURE-2 **Protocol Number:** MS201618_0034 **NCT Number:** NCT05974267 **Sponsor Name / Company:** Merck (Merck Healthcare KGaA, Darmstadt, Germany) **Investigational Product:** M5717 (Cabamiquine) and Pyronaridine (Pyronaridine tetraphosphate) **Phase of Development:** PHASE2 (Phase 2a) **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of a single dose of M5717 plus pyronaridine tetraphosphate in clearing current *Plasmodium falciparum* infection and protecting against recurrent infections in asymptomatic adults and adolescents. **Secondary Objectives:** To assess the duration of protection provided by different doses of M5717 plus pyronaridine, evaluate the additional contribution of M5717 to the duration of protection using external study data, and evaluate the pharmacokinetics (PK) of the combination.

2.2 Background & Rationale Malaria infection is a life-threatening, vector-borne disease caused by Plasmodium parasites, transmitted to humans through the bite of infected female Anopheles mosquitoes. The parasites multiply in the liver and then infect red blood cells, leading to characteristic symptoms such as cyclical fevers, chills, sweating, headache, and other flu-like manifestations. If untreated, it can progress to severe anemia, organ failure, and death, and is predominantly found in tropical and subtropical regions worldwide. Given the limited therapeutic options and the constant threat of drug resistance, there is an urgent need to discover and develop new antimalarials. This study is being conducted to explore whether the combination of M5717 (a peptide elongation factor 2 inhibitor) and pyronaridine can be effectively deployed as a single-dose chemoprevention strategy to clear asymptomatic infections and protect vulnerable populations against recurrence.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Atovaquone-Proguanil) and Dose-Comparison

Blinding: Open-label **Randomization:** Randomized **Trial Status:** COMPLETED (Study has been completed)

3.2 Planned Enrollment & Duration Targeted Enrollment: 192 participants **Number of Sites:** Multicenter (including sites in Burkina Faso, Gambia, Kenya, and Zambia) **Estimated Study Start:** 28-Nov-2023 **Estimated Primary Completion:** 29-Dec-2024

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants aged 12 to 55 years. 2. Asymptomatic *Plasmodium falciparum* malaria with no fever or other signs of acute uncomplicated malaria. 3. Microscopic confirmation using Giemsa-stained thick film, with parasitemia between ≥ 40 to $\le 10,000$ asexual parasites/μL of blood. 4. Axillary temperature $< 37.0^\circ\text{C}$ or oral/tympanic/ rectal temperature $< 37.5^\circ\text{C}$, without a history of fever during the previous 48 hours. 5. Body weight ≥ 45 kg. 6. Participants capable of giving Signed Informed consent which includes Compliance with the requirements and restriction listed in the Informed consent form. 7. Other Protocol defined Inclusion Criteria could apply.

4.2 Exclusion Criteria 1. Participants with any disease requiring chronic treatment. 2. Preplanned surgery during the study. 3. Previous treatment with pyronaridine as part of a combination therapy during the last 3 months. 4. Inadequate hematological, hepatic, or renal function as defined in the protocol. 5. Other protocol defined Exclusion Criteria could apply.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Investigational Arm:** M5717 (200 mg or 660 mg) plus pyronaridine tetraphosphate (720 mg for participants ≥ 65 kg, or 540 mg for participants 45 to <65 kg) administered as a single dose in a single-day treatment regimen. * **Comparator Arm:** Atovaquone-Proguanil (Trade Names: Malarone) 1000/400 mg once daily in a 3-day treatment regimen. (Drug Name / Drug Preferred Name: proguanil; ATC Code: P01BB01). **Route of Administration:** Oral **Duration of Treatment:** Single day (for M5717/Pyronaridine) or 3 days (for Comparator).

5.2 Concomitant & Prohibited Therapy Permitted: Routine concomitant care as required, excluding antimalarial agents not specified in the protocol. **Prohibited:** Use of other combination therapies containing pyronaridine or any off-protocol antimalarial treatment.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed consent, thick blood film microscopy, baseline labs, temperature check | | **Baseline** | Day 0 | Randomization, administration of single-dose M5717+Pyronaridine or Day 1 Atovaquone-Proguanil | | **Interim** | Variable | Safety labs, parasitemia assessments, PK blood sampling, efficacy follow-up | | **End of Study** | Follow-up | Assessment of SARS-CoV-2/Malaria recurrence, final safety evaluation, study exit |

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7. Outcome Measures (Endpoints)

7.1 Primary Outcomes Definition: Time to Parasitemia Since Negative Blood Smear after Treatment. **Measurement Method:** Microscopic evaluation of blood smears and/or PCR for parasitic clearance, evaluating the clearance of current *Plasmodium falciparum* infection and protection against recurrent infections over the specified follow-up timeframe.

7.2 Secondary Outcomes * Percentage of Participants with Parasitemia (positive blood smear). * Percentage of Participants with Polymerase Chain Reaction (PCR)-adjusted Parasitemia (Thick Smear/Microscopy, after Adjustment for Parasitemia due to new Infections as determined by Genotyping using PCR Techniques). * Percentage of Participants with PCR-adjusted Parasitemia (Thick Smear/Microscopy, after Adjustment for Parasitemia due to Recrudescence as determined by Genotyping using PCR Techniques). * Duration of protection provided by the different dosages of M5717 plus pyronaridine. * Pharmacokinetics (PK) profile of M5717 (e.g., C_{max}, AUC).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient reporting and clinical laboratory assessments. **Serious Adverse Events (SAEs):** Accelerated reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Internal sponsor mechanisms per Merck KGaA safety standards.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE/SAE assessment. **Sample Size Justification:** $N=192$ participants powered sufficiently to assess proof-of-concept differences in parasitic clearance and protection duration among treatment arms. **Handling of Missing Data:** Standard imputation methodology for continuous endpoints, with complete case analysis and appropriate regression adjustments for missing events where applicable.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and centralized remote monitoring visits by the sponsor. **Audit Trail:** Validated eCRF tracking and data clarification processes.

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11. Study Identifiers & Governance

Primary ID: MS201618_0034 **Registry Number:** NCT05974267 **Posting ID:** 604580946023048583 **Sponsor/Lead Organization:** Merck Healthcare KGaA, Darmstadt, Germany **Study Title:** Efficacy, Safety, and PK of M5717 in Combination With Pyronaridine as Chemoprevention in Adults and Adolescents With Asymptomatic Plasmodium Falciparum Infection (CAPTURE-2) **Official Regulatory Title:** Phase 2a Proof-of-Concept, Multicenter, Randomized, Open Label Study to Evaluate the Efficacy, Safety, and Pharmacokinetics of a Single Dose of the Combination M5717-pyronaridine as Chemoprevention in Asymptomatic Adults and Adolescents With Plasmodium Falciparum Malaria Infection (CAPTURE-2) **Criteria Type:** PHASE_REQUIREMENT (Trial must be in PHASE2) **Source Level:** 5

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12. Clinical Framework (Malaria Specific)

Disease Areas / Name: Malaria Infection **Preferred UMLS Name:** Asymptomatic Plasmodium falciparum infection **Preferred Name:** Resistance to falciparum malaria infection **Semantic Type:** Finding **MeSH Code:** D008279 **SNOMED ID:** 102710006 **SNOMED Hierarchy:** 186710008|Parasitic infectious disease|Parasitic infectious disease (disorder) **Diagnosis Threshold:** Microscopic confirmation (Parasitemia of 40 to 10,000 asexual parasites/μL) **Medical Methodology:** Pharmacological Chemoprevention **Optimization Target:** Effective single-dose parasitic clearance and extended protection against recurrence

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13. Study Procedures & Methodology

Management Pathway: Targeted active screening and supervised single-day or 3-day oral dosing **Education Protocol (HCP):** GCP, PK sampling procedures, and trial protocol training **Education Protocol (Patient):** Informed consent and trial compliance orientation **Quality Audit Mechanism:** Independent central and site-level source data verification

14. Observation & Follow-Up Schedule

Total Duration: Follow-up dependent on the observation of protection duration **Onsite Visit Cadence:** Intensive monitoring directly following administration (Days 1-3), scaling down to periodic follow-up **Remote Monitoring Frequency:** Intermittent centralized data review **Checklist Review Interval:** Routine reviews for parasitemia and AE tracking

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]